

Nigeria Maize Yield Intelligence

Weather-based machine learning for state-level yield analysis,
climate-stress sensitivity and explainable decision support

37 states/FCT

185 records

2020–2024

NASA POWER

3MTT Capstone Project

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From agricultural data to practical insight

A capstone focused on decision support—not just model accuracy.

THE PROBLEM

Weather variability makes maize yield analysis difficult across Nigerian states.

- Yield patterns differ across states and years.
- Weather effects are nonlinear and correlated.
- A single prediction is not enough—users need uncertainty and explanation.
- Climate scenarios must be framed carefully to avoid false forecasts.

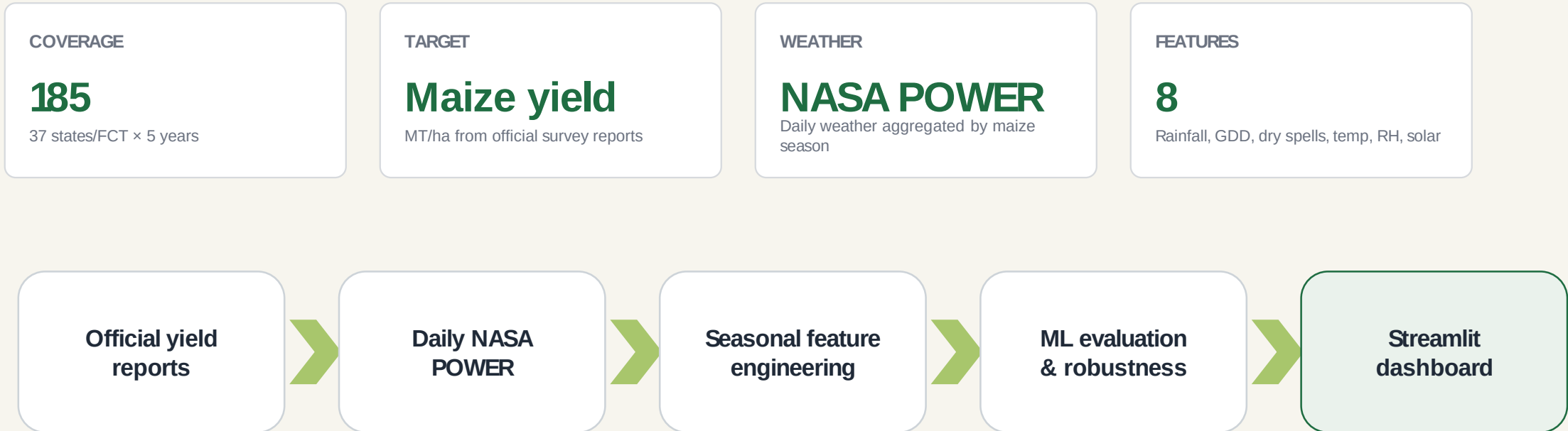
THE SOLUTION

An interactive ML dashboard combining yield, weather, robustness and explainability.

- Explore historical state-level yield and weather.
- Compare Ridge, Random Forest and LightGBM.
- Test controlled heat and rainfall stress scenarios.
- Explain predictions using held-out SHAP analysis.

Data and pipeline

A reproducible state-year panel with explicit provenance and quality control.



Dataset v10 is frozen and checksum-verified; climate-stress features are derived from the same source-of-truth weather pipeline.

The dashboard turns the research into a usable product

Five pages designed for exploration, model comparison and demonstration.

01

Overview

Dataset summary and national yield patterns

02

Yield Explorer

State-level yield, rainfall and temperature trends

03

Model Performance

Temporal, spatial and uncertainty comparison

04

Climate Stress

Interactive heat/rainfall scenario explorer

05

Explainability

Held-out SHAP feature attribution

Live-demo focus: change state, model, temperature and rainfall on the Climate Stress page and immediately show the model response.

Model performance: useful, but not overclaimed

Temporal and state-held-out tests show modest predictive skill and no clear statistical winner.

BEST 2024 MAE

0.361

LightGBM

BEST SPATIAL MAE

0.353

Random Forest

BEST 2024 R²

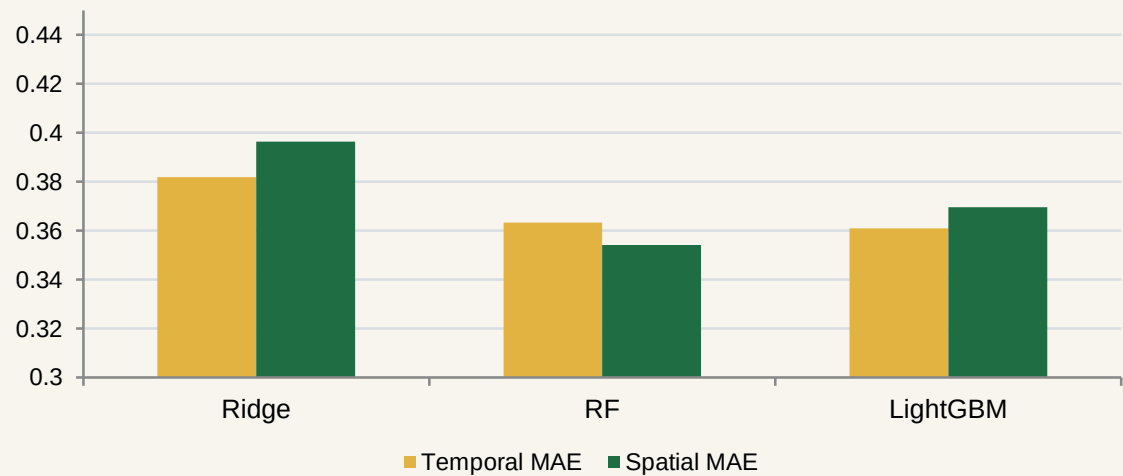
0.158

Random Forest

BEST SPATIAL R²

-0.071

Random Forest



Paired bootstrap result

No pairwise MAE difference had a 95% bootstrap confidence interval excluding zero.

Takeaway: report the lowest-error model, but avoid claiming a definitive winner.

Climate-stress explorer

Controlled daily-weather perturbations reveal model sensitivity—not future climate forecasts.

Illustrative mean predicted response

Scenario	Model	Mean change
-30% rainfall	Ridge	+0.17%
-30% rainfall	Random Forest	-1.25%
-30% rainfall	LightGBM	-2.74%
+2°C & -30% rain	Ridge	+3.49%
+2°C & -30% rain	Random Forest	-1.65%
+2°C & -30% rain	LightGBM	-2.01%

Why 04b is stronger

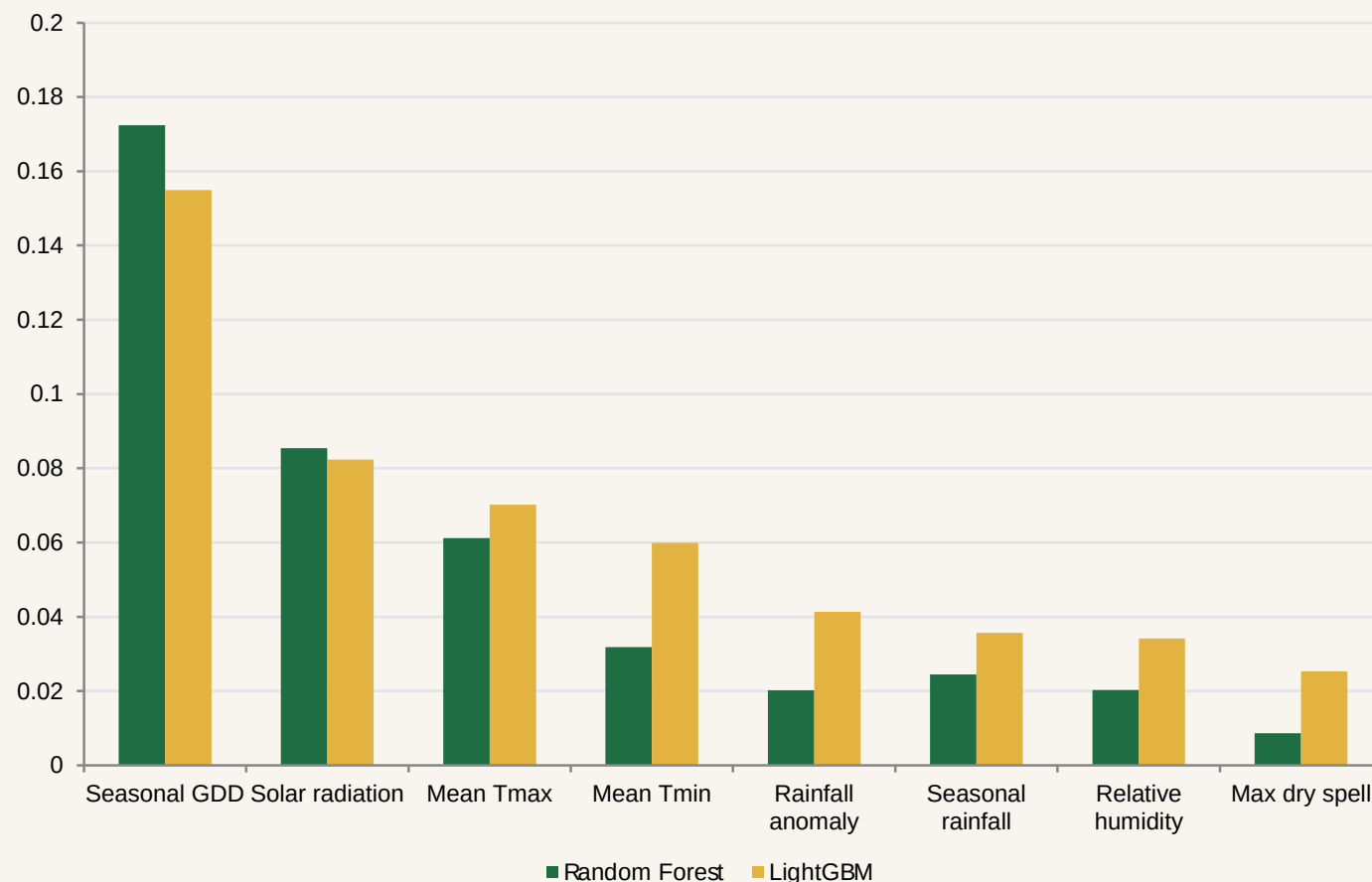
- Perturbs daily 2024 weather first
- Recomputes GDD and consecutive dry days
- Keeps 2000–2019 climatology fixed
- Preserves a clear “not a forecast” boundary

Best demo moment

Select a state → choose a model → move to +2°C / -30% rainfall → show baseline vs stressed yield.

Explainability: what the fitted models rely on

SHAP was calculated only on held-out observations and cross-checked with feature ablation.



Consistent top 3

1 Seasonal GDD

2 Solar radiation

3 Mean Tmax

Ablation also showed that removing GDD substantially reduced RF and LightGBM performance.

Interpretation rule

SHAP explains model behavior; it does not prove causality.

Reliability and limitations

The project explicitly reports uncertainty and where the models are weak.

TEMPORAL COVERAGE

89.2%

Random Forest, target 90%

SPATIAL COVERAGE

89.7%

Random Forest, target 90%

TEMPORAL WIDTH

65.3%

of mean observed yield

SPATIAL WIDTH

74.6%

of mean observed yield

What works

- Reproducible data + checksums
- Held-out temporal and spatial evaluation
- Conformal prediction intervals
- Climate-stress identity and method checks

What remains limited

- Only 185 state-year observations
- Weather-only models omit management, soil and economics
- Spatial R^2 remains weak/negative
- Stress scenarios are controlled perturbations, not projections

What this capstone delivers

1

Interactive product

A Streamlit dashboard for state-level exploration and live scenario testing.

2

Applied ML

Ridge, Random Forest and LightGBM with temporal and spatial evaluation.

3

Responsible interpretation

Uncertainty, explainability and explicit limits on climate claims.

4

Research-ready foundation

The same pipeline supports a follow-on academic paper after capstone submission.

Thank you

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